# Use-Case Flow Specification

UC-01 - Submit Lost Luggage Claim

| Name | Mohammed Zahoor Mashahir | SRN | PES1UG24CS579 | Section | J |
|---|---|---|---|---|---|

| | |
|---|---|
| **Primary Actor** | Passenger |
| **Supporting Actors** | Baggage Service Agent; Airport Baggage Scan Network |
| **Trigger** | A passenger's checked bag is not delivered at the arrival carousel. |
| **Preconditions** | Passenger can provide the booking reference and bag-tag number; the bag is not delivered; no active duplicate claim exists for the same passenger and bag. |
| **Postconditions** | Success: claim ID, matched scan timeline, recovery alert, and agent task are stored. Minimum guarantee: if no scan matches, the claim remains stored as 'Under Search' and is queued for agent investigation. |

## Main Success Scenario

1. Passenger selects 'Report Lost Luggage' in the portal.
2. System requests booking reference, passenger surname, and bag-tag number.
3. Passenger enters the identifiers, itinerary, contact details, bag description, delivery address, and optional supporting images.
4. System validates booking ownership, bag-tag format, mandatory fields, and absence of an active duplicate claim.
5. System displays the completed claim for review; Passenger confirms the declaration and submits it.
6. System creates a unique claim ID, timestamps the claim, and sets status to 'Submitted'.
7. System cross-references the report against unrouted scan logs from all connected terminals and transit hubs.
8. System links the matching bag timeline, sets status to 'Located - Recovery in Progress', generates a recovery alert, and assigns an agent task.
9. System displays the claim ID and status and sends the Passenger a tracking link. The use case ends successfully.

## Alternate Flow A - No Matching Scan Found at Step 8

- System sets the claim status to 'Under Search' and assigns it to the Baggage Service Agent queue.
- System subscribes the claim to subsequent scan events so new matches are checked against the bag tag and itinerary.
- System informs the Passenger that no current scan match was found, displays the claim ID and tracking link, and ends the use case with the claim active.